

Zulay C. Rodriguez

M.S. Ecology and Evolutionary Biology | B.S. Biological Sciences

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Education

June 2021-April 2023	<i>M.S., Ecology and Evolutionary Biology</i> University of Michigan GPA: 3.90 Relevant coursework: Introduction to EEB Reservation, Teaching College Science, R-coding
August 2018- December 2020	<i>B.S., Biology (Chemistry minor)</i> Florida State University GPA: 3.82; Dean's List for all semesters Honors in the major Relevant coursework: Organic Chemistry I/II, Biology I/II with lab, Genetics, Eukaryotic Diversity, Evolution, Ecology, and Animal behavior.

Awards & Fellowships

August 2022	Certificate of Completion from the Association of Colombian Herpetology
March 2022	Research Award - Block Grant from the Ecology and Evolutionary Department at the University of Michigan
January 2021	Certificate of Appreciation from the Ecological Society of America Student Services Program
December 2020	Certificate of Completion from the Florida State Honors Program
June 2020	Certificate of Participation from LSAMP-Brookhaven National Lab Survey of Scientific Computing Workshop
May 2020	Certificate of Completion from Florida-Georgia Louis Stokes Alliance for Minority Participation
August 2018	Florida Bright Futures Scholar Recipient
March 2018	My Hero Nomination from PRIDE club at Miami Dade College

Teaching Experience

August 2021-April 2023

Graduate Student Instructor

Lead and manage either a discussion or lab-based course. Engage students in course work by exploring new ways to communicate and understand science. Additionally, encourage and mentor students through their journey in STEM.

December 2022

Guest Lecturer

Collaborated with a professor to create and present a lecture. Created a presentation that broke down my master's thesis research to a class of undergraduates. Lectured students on my journey in science as well as the basics of mimicry in snakes.

January 2021-April 2021

Teacher Assistant

Taught introductory biology for nonmajors at Florida State University. Managed two online lab sections of the course. Responsibilities included grading, managing zoom teaching sessions, conducting lab activities, and encouraging student engagement.

January 2019-April 2021

Student Athlete Tutor

Managed several student athletes at once. Introduced new study skills and methods to promote academic success. Taught various subjects ranging from introduction to statistics, general biology, and Spanish. Maintained a detailed report of student progress.

Research Experience

December 2021-April 2023

Davis-Rabosky Lab: Master's Thesis Research

I identified a gap in the understanding of mimicry in coral snakes. I then used museum collections, CT scans, and 3-dimensional models to collect data on fang and skull morphology of models, mimics, and non-mimics. Utilized data analysis skills and R coding to conduct spatial analysis on the distribution of mimicry and garner evidence for what processes are maintaining mimicry in coral snake mimicry systems.

June 2021-July 2021

University of Michigan: Frontiers Field Work

Navigated several field problems and practiced several skills including executing various data collection methods, statistical analysis in R, and presenting findings. Additionally, I designed, executed, and presented my own experiment and findings on factors involved in habitat selection for *Camponotus Pennsylvanicus* (black carpenter ants).

January 2021-May 2021

Florida State University: Honors in the Major

Directed my own research project under the guidance of Dr. Thomas E. Miller from Florida State University. Collected and then analyzed field data. Practiced troubleshooting skills and creatively overcoming any obstacles. Produced and defended a thesis on *Peucetia viridans* (green lynx spider).

August 2019- May 2020

Directed Individual Study

Aided Dr. Kim Hughes at Florida State University in her investigation of the mechanism which maintains genetic diversity in natural populations through quantification of male color patterns in Trinidadian guppies using multivariate geometric morphometrics.

May 2016-August 2016

Florida International University Summer Research Internship

Ran an independent experiment under Dr. Sian Evans and investigated the effects of diet on interfamilial relationships in Owl Monkeys.

Publications

February 2022

Understanding spatially complex cryptic coloration in the green lynx spider *Peucetia viridans*

Published in Ecological Entomology this article describes a novel method for quantifying color variation and its application in identifying color-matching (cryptic coloration) in natural populations of green lynx spiders.

December 2021

Dressing right for the occasion: camouflage in spiders

EEBlog article summarizing the findings of my undergraduate research on cryptic coloration in green lynx spiders (*P. viridans*).

Presentations

March 2023

Latinx Research Week at the University of Michigan

Presented some of my preliminary findings with other members of the natural science community that had a Latinx identity. I engaged in conversation that discussed Latinx excellence and engagement in STEM. As well as educated the general public on coral snake mimicry and evolutionary aspects of mimicry.

August 2022

Colombian Congress of Herpetology at ICESI University

Traveled to Colombia and shared the preliminary findings of my master's thesis project. Created and presented a bilingual presentation titled “Mantener el Exceso Mimético: una Paradoja Evolutiva en el Mimetismo de la Serpiente de Coral (Maintaining Excess Mimicry: An Evolutionary Paradox in Coral Snakes)” which discussed the conundrum of excess mimics within a batesian system in coral snakes.

April 2020

Undergraduate Research Symposium at Florida State University.

Presented my undergraduate honors project titled “Do Predator Spiders Change Color to Hide in Flowers” which discussed the preliminary data regarding the capacity of green lynx spiders (*P. viridans*) to display active cryptic coloration.

Service and Outreach

March 2022 & 2023

Feria de Ciencias

Science fair for Spanish speaking elementary students. Created and presented a science demonstration with other graduate students to encourage student interest and curiosity in ecology and evolution.

September 2019-May 2020

Strategies for Ecology Education, Diversity and Sustainability program (SEEDS)

Focuses on connecting upperclassmen with lower classmen of various backgrounds in order to provide guidance about navigating ecology as an undergraduate. This is a program within the Ecological Society of America.

September 2019-January 2021

Florida Georgia Louis Stokes Alliance for Minority (FGLSAMP)

A coalition of colleges and universities which aid minorities in undergrad to complete their STEM degrees and apply to graduate school. Involved working in collaboration with local high schools to encourage minority participation in STEM.

September 2018-January 2021

Cuban American Student Association (CASA)

Formed to recognize academic excellence among Cuban students and provide leadership opportunities aimed to give back to the Latinx community.